LaB 08

Lucas Rocha

```
import polars as pl

caminho = "C:/Users/User/Downloads/"

basics = pl.read_csv(
    caminho + "title.basics0.tsv.gz",
    separator="\t",
    null_values=r"\N",
    quote_char=None,
)
ratings = pl.read_csv(
    caminho + "title.ratings.tsv.gz",
    separator="\t",
    null_values=r"\N",
    quote_char=None,
)
principals = pl.read_csv(
    caminho + "title.principals0.tsv.gz",
    separator="\t",
    null_values=r"\N",
    quote_char=None,
)
```

1 Quais são os 5 filmes com as maiores notas (averageRating)? Apresente uma solução capaz de desempatar os filmes baseando-se no número de votos recebidos.

```
q1 = (
    basics.filter(pl.col("titleType") == "movie")
    .join(ratings, on="tconst", how="inner")
    .sort(by=["averageRating", "numVotes"], descending=[True, True])
    .select(["tconst", "primaryTitle", "averageRating", "numVotes"])
    .head(5)
)

print(q1)
```

```
shape: (5, 4)
┌────────────┬──────────────────────────────┬─────────────────┬────────────┐
│ tconst     ┆ primaryTitle                 ┆ averageRating ┆ numVotes  │
│ ---        ┆ ---                          ┆ ---             ┆ ---        │
│ str        ┆ str                          ┆ f64             ┆ i64        │
╞════════════╪══════════════════════════════╪═════════════════╪════════════╡
│ tt33075815 ┆ Kaveri                       ┆ 10.0            ┆ 1023       │
│ tt27815015 ┆ Kurukku                      ┆ 10.0            ┆ 451        │
│ tt28450376 ┆ Jedal Dar Omghe 30 Metri     ┆ 10.0            ┆ 142        │
│ tt27859713 ┆ Sargashte                    ┆ 10.0            ┆ 134        │
│ tt33507675 ┆ Gorgeous Rascal              ┆ 10.0            ┆ 115        │
└────────────┴──────────────────────────────┴─────────────────┴────────────┘
```

2.Qual é o gênero mais frequente entre os filmes com nota maior que 8?

```
q2 = (
    basics.filter(pl.col("titleType") == "movie")
    .join(ratings, on="tconst", how="inner")
    .filter(pl.col("averageRating") > 8)
    .drop_nulls(subset=["genres"])
    .with_columns(pl.col("genres").str.split(","))
    .explode("genres")
    .group_by("genres")
    .agg(pl.len().alias("frequencia"))
    .sort(by="frequencia", descending=True)
    .head(1)
)

print(q2)
```

```
shape: (1, 2)
┌──────────────┬──────────────┐
│ genres       ┆ frequencia │
│ ---          ┆ ---          │
│ str          ┆ u32          │
╞══════════════╪══════════════╡
│ Documentary ┆ 10564        │
└──────────────┴──────────────┘
```

3.Quais são os 3 atores/atrizes que mais participaram de filmes com nota maior que 7.5?

```
q3 = (
    basics.filter(pl.col("titleType") == "movie")
    .join(ratings, on="tconst", how="inner")
    .filter(pl.col("averageRating") > 7.5)
    .join(
        principals.filter(pl.col("category").is_in(["actor", "actress"])),
        on="tconst",
        how="inner",
    )
    .group_by("nconst")
    .agg(pl.len().alias("total_filmes"))
    .sort(by="total_filmes", descending=True)
    .head(3)
)

print(q3)
```

```
shape: (3, 2)
┌────────────┬──────────────┐
│ nconst     ┆ total_filmes │
│ ---        ┆ ---          │
│ str        ┆ u32          │
╞════════════╪══════════════╡
│ nm0004660 ┆ 231          │
│ nm0595934 ┆ 155          │
│ nm3183374 ┆ 124          │
└────────────┴──────────────┘
```

```
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()
```

```
[1] "2026-10-14 18:05:45 -03"
```

```

```